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Unleashing human potential: An artificial intelligence competency framework for K–12 education

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ABSTRACT

This study explores strategic approaches for integrating artificial intelligence (AI) into K–12 education to prepare learners for the AI era. It addresses two critical questions: 1) What are the key components of AI competency frameworks identified in current research? 2) What framework can guide the effective, responsible integration of AI in K–12 while prioritising human values? We conducted a scoping review of 54 studies, identifying three core student competencies: foundational AI knowledge, practical and cognitive skills of using AI, and ethical awareness. However, existing frameworks place limited emphasis on developing human values. Consequently, we propose a three-component developmental framework—Understanding, Using, and Unleashing—to guide AI integration. The 'Understanding' component cultivates a conceptual understanding of AI, while 'Using' emphasizes practical and cognitive skills of using AI to enhance learning. Finally, 'Unleashing' highlights AI's potential to empower personal growth, and fostering a spiritual self capable of making value-based decisions and do good. This framework aims to prepare learners to contribute meaningfully to future society. Schools must teach students to use AI for empowerment and good, avoiding mere reliance or shortcuts. Through practical activities, education should guide learners to unleash their potential and harness technology for spiritual self-development.

1. Introduction

Does the widespread use of artificial intelligence (AI) in our societies risk diminishing the unique roles and values of human? The rapid advancement of AI technologies, particularly the rise of generative AI tools, is reshaping how decisions are made in human society (Brandao, 2025; Stone et al., 2020). Unlike traditional technologies, which primarily served to enhance productivity and reduce physical labour, AI introduces a fundamental shift. By automating decisions on a massive scale, this technology moves beyond the role of a tool and begins to replace human decision-making in areas ranging from creative industries to routine workforce tasks (Corvello, 2025; Taeihagh, 2025). By automating "cognitive activities", AI has sparked widespread reflection and concern (Gruenhagen et al., 2024; Steyvers & Kumar, 2024). Many perceive this as a threat to the unique value of human abilities, as tasks once considered exclusive to human intelligence—such as problem-solving, creativity, and ethical reasoning—are increasingly performed by machines (Gerlich, 2025; Lee et al., 2025; Schmager et al., 2025). This raises profound questions: in a society increasingly shaped by intelligent machines, what roles will remain uniquely human?

This study addresses this issue from the perspective of education and particularly in K–12 education. AI is now changing the landscape of primary and secondary school education. A growing body of literature explores how AI can be used to improve teaching efficiency, automate feedback, or support personalised learning (Banihashem et al., 2024; Gnanaprakasam & Lourdusamy, 2024; Holmes et al., 2019; Vorobyeva et al., 2025; Zawacki-Richter et al., 2019). A crucial issue raises: Integrate these powerful technologies into school education in a way that enhances human potential, upholds core human values, and ensures that education motivates the next generation to grow into better individuals, rather than losing their drive for self-improvement in the shadow of AI.

As Kong (2025) argues, achieving robust AI competency necessitates addressing those concerns. However, current frameworks, such as TPACK (Kim, Jang, Choi, et al., 2021) and SAMR (Sirotek et al., 2025),

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while valuable for outlining digital integration, do not fully encompass the critical dimensions of human values and human potential. Recently, there has been an increased emphasis on learner-centered approaches in AI education, which prioritize strategies that foster learner development. This includes learner-centered feedback mechanisms (Aldino et al., 2025), targeted interventions to enhance students' metacognitive skills (Li et al., 2025; Siddiqui et al., 2025; Yan et al., 2025), promoting harmonious coexistence among human and non-human communities (Global Harwell Collaboration, 2025), and building resilience to navigate AI while preserving independent thinking (Raian, 2025). Drawing on these preliminary studies, it is crucial to rethink an AI competency framework that prominently emphasizes human values and human potential.

2. Background of study

2.1. AI literacy and AI competency

Contemporary discussions of artificial intelligence in education are closely related to two intertwined concepts: AI literacy and AI competency. Scholars widely regard AI literacy/competency as an essential scaffold enabling students, teachers, and broader stakeholders to understand and respond to AI technologies. While AI literacy focuses on understanding—knowledge of concepts and skills—AI competency emphasizes the effective application of that knowledge in meaningful and beneficial ways, encompassing both confidence and proficiency (Chiu, Ahmad, et al., 2024). Long and Magerko (2020), for instance, proposed a framework of seventeen competencies across four categories, ranging from conceptual understanding and functional knowledge to the evaluation of social impacts. Their work positions AI literacy as the basis for critical decision-making and human–AI collaboration. Similarly, Casal-Otero et al. (2023), in a systematic review, identified two dominant strands in AI literacy research: technical (e.g., programming, data handling) and theoretical (e.g., conceptual understanding, ethics). As seen in Kong et al.'s (2024) framework for K–12 and Su et al.'s (2023) early childhood–oriented designs, the trend is clear: AI literacy is evolving from a specialized skill into a core competency embedded across different components of education.

While AI literacy and AI competency often refers to the foundational knowledge and understanding required to engage with AI, practical and cognitive skills and higher-order capacities needed to effectively and ethically work with AI systems, it is urgent that we reimagine what it means for human beings to grow, reflect, and flourish in an AI-mediated world, by integrating essential humanistic and philosophical perspectives into AI education.

2.2. The potential and challenges of AI for human development in K–12 education

On the one hand, AI propel human development through augmented efficiency and inspiration (Calleja & Camilleri, 2025; Huang, 2024). On the other hand, it risks precipitating declines in intrinsic creativity, mental agility, and personal goal orientation. For example, Zhai et al. (2024) caution that excessive dependence on AI dialogue systems can erode independent thinking. Nikolic et al. (2023) delineate threats to academic assessment integrity, noting AI's proficiency in producing credible responses to sophisticated queries. Similarly, cognitive debt has been observed in learners using AI assistants without guidance, resulting in weaker neural connectivity (Kosmyna et al., 2025). As a result, it is urgent to nurture authentic self-development, transcending mere technological integration to encompass cognitive, ethical, and existential dimensions (Kong, 2025).

This research aims to answer the following two questions: 1) What are the key components of AI competency frameworks as identified in current research on K–12 education? 2) What framework can be proposed to guide the effective and responsible integration of AI in K–12 education for preparing learners to contribute to the future society while prioritising human values?

3. Methodology

3.1. Research design

To address the research question—"What are the key components of AI competency frameworks as identified in current research on K–12 education?"—this study adopted a scoping review methodology. This approach is particularly suitable for mapping the breadth and depth of available evidence and identifying key concepts in the rapidly emerging field of AI education (Arksey & O'Malley, 2005; Tricco et al., 2018). The review was guided by the methodological framework outlined by Arksey and O'Malley (2005), which includes five stages: identifying the research question, searching for relevant studies, selecting studies, charting the data, and collating, summarizing, and reporting the results. Reporting followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR) guidelines (Tricco et al., 2018). No review protocol was registered. Consistent with standard practice in scoping reviews, critical appraisal of the methodological quality of individual sources was not performed, as the primary objective was to comprehensively map existing AI competency frameworks components in K–12 education rather than evaluate study rigor (Tricco et al., 2018).

3.2. Search strategy

As Fig. 1 shows, search was conducted in three major databases covering education and interdisciplinary research: Scopus, Web of Science Core Collection, and ERIC. Each database was searched using Boolean operators to combine controlled descriptors (where available) and free-text keywords. Searches were limited to peer-reviewed journal articles and reviews published in English between 2021 and 2025.

The search was conducted on the first of September 2025. The following search strings were used (database-specific syntax was applied): (1) Scopus: TITLE-ABS-KEY(("artificial intelligence" OR AI) AND ("K–12″ OR "primary school*" OR "secondary school*" OR "elementary education") AND (literacy OR competency* OR knowledge OR awareness)); (2) Web of Science (Core Collection): TS=(("artificial intelligence" OR AI) AND ("K–12″ OR "primary school*" OR "elementary education" OR "secondary school*") AND (literacy OR competency* OR knowledge OR awareness)); (3) ERIC: ("artificial intelligence" OR AI) AND (literacy OR competency* OR awareness OR "AI literacy") AND ("K–12″ OR "primary education" OR "secondary education" OR "elementary school" OR "high school"). Fig. 1 shows the PRISMA flow chart and the outcomes of the search. The asterisk (*) were used as a wildcard to include variations of words, such as plural forms or different endings, in order to broaden the search scope and ensure comprehensive results.

4. Study the characteristics of reviewed articles

4.1. Classification 1: research on AI concept understanding

This section includes research focused on students' foundational knowledge of AI systems, including data and algorithmic literacy, as well as an understanding of AI's capabilities and limitations. Table 1 shows fifteen research studies on data and algorithmic literacy from 2021 to 2025. Research on the "Understanding dimension" of K–12 AI competency centres on students' foundational knowledge of AI concepts, from basic principles to the intricate workings of algorithms and data. This body of work reveals a shift from simply defining AI to exploring how students conceptually grasp its inner mechanisms.

A key trend in this research is the exploration of how to make complex AI concepts accessible to young learners. This approach helps

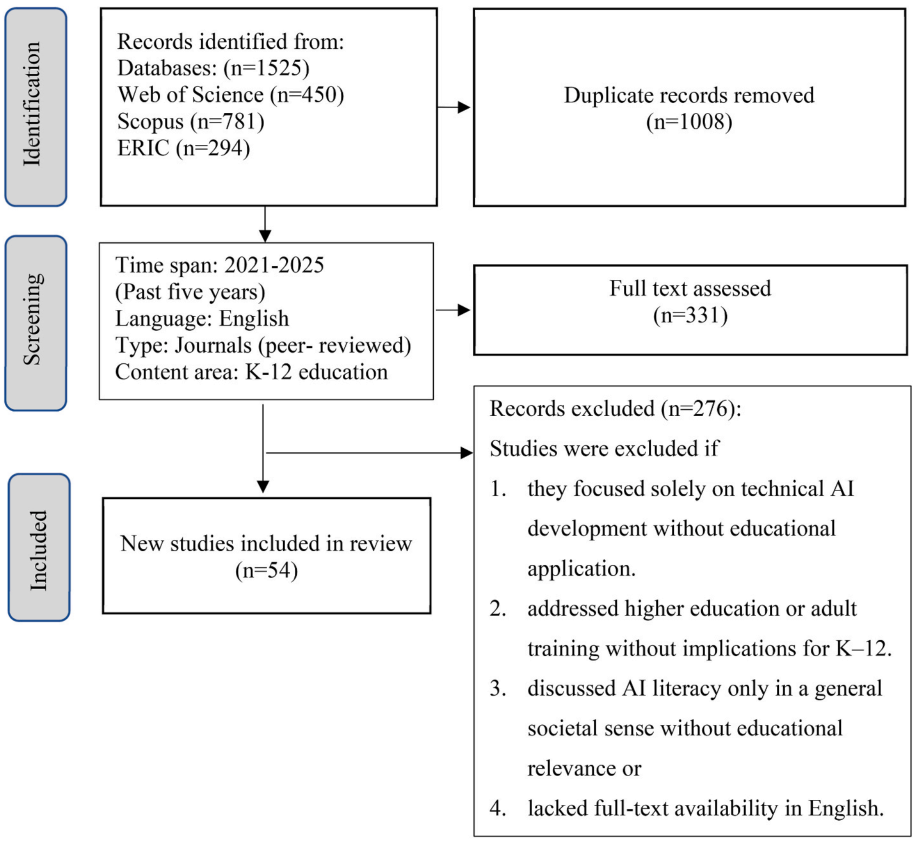

Fig. 1. The PRISMA flow chart and the outcomes of the search.

Table 1
Fifteen research studies on data and algorithmic literacy, 2021–2025.

Author (Year)	Study Focus	Methodology	Key Findings
Chai et al. (2021)	Primary students' perceptions and intentions to learn AI	Survey questionnaire	Students' attitudes towards AI, built on their conceptual understanding, are a major factor in their learning intentions.
Chee et al. (2025)	A comprehensive competency framework for AI literacy	Framework development	Proposes a multi-level AI literacy framework, where conceptual understanding is a foundational component for different learner groups.
Chiu, T. K. (2021)	Holistic design of K–12 AI education curriculum	Thematic analysis	Proposes a curriculum design model emphasizing the importance of teaching core AI concepts.
Huang, X. (2021)	Cultivating students' key competencies via AI education in China	Literature review, framework analysis	Identifies core AI competencies, with foundational knowledge of data and algorithms as a primary component.
Kim, Jang, Choi, et al. (2021)	Analysing teacher competency for K–12 AI education	TPACK framework analysis	Teachers need strong technological and content knowledge of AI to effectively teach foundational concepts to students.
Kim, Jang, Kim, et al. (2021)	Designing an AI curriculum for elementary school	Curriculum development	Outlines a curriculum for elementary students, emphasizing the “why” and “what” of teaching core AI concepts at this level.
Kong et al. (2023)	Conceptualizing AI literacy	Exploratory literature review	Defines AI literacy as a composite of knowledge, skills, and attitudes, with an emphasis on conceptual understanding of AI's core principles.
Steinbauer et al. (2021)	Differentiated discussion of K–12 AI education	Review, discussion	emphasis sound knowledge about principles and concepts of AI.
Tedre et al. (2021)	Teaching machine learning in K–12 classrooms	Pedagogical review	Stresses the importance of tailoring AI concepts to different age groups and educational levels within the K–12 system.
Touretzky and Gardner-McCune (2022)	The concept of “AI thinking” for K–12	Theoretical framework	Defines “AI thinking” as a crucial competency that involves understanding how AI systems perceive, reason, and learn.
Wu et al. (2024)	Classroom instruction on K–12 AI education	LLMs-based analysis and manual analysis.	Analyses how AI literacy, including conceptual understanding, is taught in classroom settings using various pedagogies and tools.
Yu, W. (2025)	A conceptual framework for AI literacy	Framework development	Proposes a new competency-focused framework for AI literacy, with a strong emphasis on foundational knowledge.
Yim, I. H. Y. (2024)	AI literacy framework for primary school education	Systematic Review and analysis	Develops a new framework specifically for primary education, highlighting core concepts and a need for more research on younger learners.
Zhong and Liu (2025)	Evaluating AI literacy of secondary students	Framework and scale development	Creates a tool to measure secondary students' AI literacy, with conceptual understanding as a key dimension for assessment.
Zhou et al. (2025)	Defining, enhancing, and assessing AI literacy and competency	Systematic review and Meta-Analyses	Synthesizes existing definitions and suggests a modified framework where conceptual understanding is the basis for higher-level competencies.

learners avoid the anthropomorphic trap of attributing human-like qualities to algorithms, fostering a more accurate and critical understanding of these technologies. The studies emphasize teaching foundational concepts of data science and algorithmic intelligence, enabling students to build a solid understanding of machine learning and deep learning by first grasping essential terms like tokens and embeddings. Studies by Chiu (2021) and Kim et al. (2021) provide frameworks for K–12 AI curricula, emphasizing the importance of foundational knowledge. Moving beyond theoretical definitions, papers by Tedre et al. (2021) and Steinbauer et al. (2021) highlight pedagogical methods for teaching complex machine learning and algorithmic concepts. More recent studies, such as those by Simbeck and Kalff (2024) and Yim (2024), provide frameworks for teaching these concepts specifically in elementary education.

Furthermore, a significant sub-theme is the focus on the capabilities and limitations of AI systems. These studies also emphasize the boundaries between artificial intelligence and human intelligence, aiming to dispel misconceptions of AI as magical or anthropomorphic. Table 2 shows three research studies focused in these areas from 2021 to 2025. Studies by Ali et al. (2021) and Henry et al. (2021) explore how children's perceptions of AI are shaped through engaging activities like games and role-playing, which challenge their preconceptions and highlight the boundaries of AI's "intelligence". Druga and Ko (2021) found that hands-on coding activities can directly influence a child's understanding of AI, helping them realize that AI is not magical but is limited by its training data and programming. Collectively, this research demonstrates that building K–12 AI literacy begins with a strong conceptual foundation, empowering students to understand AI's core principles, its capabilities, and its inherent limitations.

4.2. *Classification 2: research focusing on the practical and cognitive skills aspect of AI competency*

Research into the 'Using dimension" suggests that using AI is not a single skill but a layered practice—one that must be grounded in a clear understanding of AI's strengths and limitations. Effective use requires deliberate design, institutional support, and flexible approaches such as strategic, comparative, reflective, and critical engagement.

Table 3 shows sixteen research studies focused on the critical use of AI tools from 2021 to 2025. This body of literature shifts the focus from foundational knowledge to the practical and cognitive skills needed to navigate and leverage AI technologies effectively. A significant trend is the development of critical use competencies. Several reviews, including those by Casal-Otero et al. (2023) and Tan and Tang (2025), emphasize the importance of teaching students to evaluate AI-generated content, identify biases, and verify information. Studies on generative AI, like

Table 2
Three research studies focused on AI system capabilities and limitations, 2021-2025.

Author (year)	Study Focus	Methodology	Key Findings
Ali et al. (2021)	Children's perceptions of AI's role in society	Exploratory study	Explores how children view AI's capabilities and its role as a tool or partner, touching on their conceptual awareness of what AI can and cannot do.
Druga and Ko (2021)	Children's perceptions of machine intelligence through coding	Exploratory study	Investigates how hands-on coding activities affect children's perceptions of machine intelligence, demonstrating a deeper understanding of AI's limitations.
Henry et al. (2021)	Teaching AI through a role-playing game	Design-based research (DBR) approach	Uses a game to challenge students' pre-existing notions of AI, compelling them to consider the boundaries of intelligence and AI's capabilities.

Table 3
Sixteen research studies focused on the critical use of AI tools, 2021-2025.

Author (Year)	Study Focus	Methodology	Key Findings
Adisa and Adefisayo (2025)	Middle school students' perspectives on adopting generative AI	Phenomenological research design	Students are aware of generative AI tools but need guidance on critical evaluation of their outputs.
Alé et al. (2025)	Scientific practices for using and creating with AI in K–12	Scoping review	Highlights the need for students to critically interpret AI-generated scientific insights.
Casal-Otero et al. (2023)	A systematic literature review of AI literacy in K–12	Systematic review	Synthesizes research on AI literacy, emphasizing independent thinking as a key skill.
Cheah et al. (2025)	Teachers' preparedness for integrating generative AI in K–12	Cross-sectional survey research method	Examines teachers' readiness for using generative AI, which includes their critical understanding of its challenges and biases.
Cheung et al. (2025)	Epistemic insights of AI in science education	Systematic review	Explores the application of AI in science education, touching on the critical skills required to interpret AI-generated scientific insights.
Klar (2025)	Students' effective use of generative AI	Experimental, value-added research design	Argues that while generative AI is easy to use, students struggle with using it effectively and critically.
Lee and Kwon (2024)	A systematic review of AI education in K–12 classrooms	Systematic review	Analyses teaching strategies, many of which are designed to cultivate students' skills in using AI critically.
Li et al. (2024)	Learning task design for K–12 AI education	Systematic review	Examines how tasks are designed to develop practical and critical skills in using AI.
Liu and Zhong (2024)	How educators teach AI in K–12 education	Systematic review	Explores how teachers instruct students on the practical and critical use of AI.
Marzano (2025)	Generative AI in K–12 teaching and learning	Systematic review	Examines the use of generative AI with a focus on its practical application and the critical skills required.
Sanusi, Olaleye, Oyelere, and Dixon (2022)	Learners' competencies for AI education in Africa	Survey study	Examine students' competencies, including their ability to critically apply AI knowledge in practical settings.
Su et al. (2024)	Teaching AI in K–12 classrooms	Scoping review	Covers pedagogical practices that aim to develop students' practical and critical skills in AI.
Tan and Tang (2025)	AI literacy in K–12 education	Systematic review	Focuses on how AI skills, including

(*continued on next page*)

Table 3 (*continued*)

Author (Year)	Study Focus	Methodology	Key Findings
			critical use, are taught and assessed in empirical studies.
Wang and Lester (2023)	K–12 education in the age of AI	Theoretical paper	Advocates for a comprehensive approach to AI literacy that includes the critical use of AI tools.
Wang et al. (2023)	Measuring user competence in using AI	Scale development	Focuses on developing a tool to measure users' practical and critical competence in using AI.
Yang (2025)	AI-supported L2 vocabulary acquisition	Systematic review	Synthesizes research on AI support on vocabulary acquisition, emphasizing its usage for language learning

Klar (2025), highlight that while students find these tools easy to operate, they struggle with using them effectively and with a critical eye. This points to a clear need for pedagogical approaches that cultivate skills beyond simple operation. The competence of teachers in this area is also a key concern, as evidenced by Cheah et al. (2025), who examined teachers' preparedness to integrate these new technologies.

Equally important is the focus on creative and strategic use. Table 4 shows eleven research studies focused on strategical and creative use of AI tools from 2021 to 2025. Many studies move beyond general AI literacy to explore its application in specific contexts. For example, Ye et al. (2025) and Wen et al. (2025) illustrate the strategic use of AI to enhance specific learning processes, such as collaborative learning or vocabulary acquisition. Other research, such as that by Hsu and Hsu (2025) and Ng et al. (2022), highlights how AI can be a powerful tool for creative expression, such as in generating drawings or writing digital stories. This body of work underscores a pedagogical shift from viewing AI as a subject to be learned to a tool for fostering creativity and problem-solving.

Collectively, these studies demonstrate that effective AI use in K–12 is multifaceted. It requires a blend of critical evaluation and strategic application, empowering both students and teachers to harness AI not just as a source of information but as a versatile tool for learning, creation, and innovation.

4.3. Classification 3: research focusing on using AI for developing metacognitive insight, emotional resilience, and ethical awareness

This category of research explores higher-order dimensions of AI competency, shifting the focus from understanding and using AI tools to examining how students develop metacognitive insight, emotional resilience, and ethical awareness in AI-mediated learning contexts. Rather than treating AI as merely a tool, these studies emphasize the unleash of students' own potential—intellectual, affective, and moral—through reflective and value-driven engagement with technology. We name this dimension as "Unleashing dimension" and shall be elaborate more in section 5.

Table 5 presents four research studies on learning affect and metacognitive capabilities conducted between 2021 and 2025. For instance, Chiu, Moorhouse, et al. (2024) examined how teacher support can significantly boost students' motivation to learn with AI chatbots. At a more advanced level, Kong et al. (2023) evaluated a program designed to enhance students' self-efficacy and ethical awareness, showing that confidence in using AI and a sense of responsibility is deeply

Table 4
Eleven research studies focused on strategical and creative use of AI tools, 2021-2025.

Author (Year)	Study Focus	Methodology	Key Findings
Almuhanna (2025)	Teachers' perspectives on using AI for learning materials	Qualitative study	Explores how teachers creatively and strategically use AI to enhance educational resources.
Cheah and Kim (2025)	STEM teachers' perceptions of generative AI integration	Mixed-methods survey design	Examine how STEM teachers use generative AI to support their creative teaching practices.
Hsu and Hsu (2025)	Teaching AI with games and generative AI drawing	Quantitative study	Explores how using AI for creative purposes, like generating drawings, can foster computational thinking.
Kahila et al. (2024)	A pedagogical framework for cultivating data agency and creativity	Framework development	Proposes a framework for fostering children's creative abilities and strategic use of AI.
Kim and Kwon (2024)	Tangible tools in AI education	Mixed Method Study	Discusses how tangible tools are used to strategically teach AI concepts to elementary students.
Kong and Yang (2024)	Human-centered framework for using generative AI	Framework development	Introduces a framework for using AI as a strategic tool to promote self-regulated learning.
Ng et al. (2022)	Using digital story writing to develop AI literacy	Mixed Method Study	Highlights a pedagogical approach that uses AI as a creative tool for digital storytelling.
Sanusi, Olaleye, Agbo, and Chiu (2022)	Learners' competencies in AI education	Survey Study	Discusses a wide range of learner competencies, including the practical and strategic skills required for AI application.
Wen et al. (2025)	AI-powered vocabulary learning for primary school students	Experimental study	Investigates the strategic use of an AI tool to facilitate vocabulary acquisition in a specific educational context.
Ye et al. (2025)	Synergizing AI and collaborative learning for Chinese characters	Experimental study	Explores the strategic use of AI to support collaborative learning in a specific subject.
Yue et al. (2025)	A pedagogical framework for students as AI literate designers	Framework development	Proposes a framework where students are taught to use AI creatively as a design tool.

intertwined. Zhao et al. (2022) further reinforced this by focusing on how teachers' self-efficacy directly impacts their ability to teach AI literacy effectively.

Another central theme is the development of AI ethical awareness. Table 6 presents six research studies on AI ethical awareness conducted between 2021 and 2025. As shown in reviews by Ma et al. (2025) and Ng et al. (2024), a key goal of modern AI education is to foster a sense of responsible use. Studies by Williams et al. (2023) and Zhang et al. (2023) provide practical insights from curriculum design, demonstrating how ethical discussions around bias, privacy, and societal impact can be woven into technical learning for middle and high school students. This is complemented by theoretical work from Adams et al.

Table 5
Four research studies focused on learning affect and metacognitive capabilities, 2021-2025.

Author (Year)	Study Focus	Methodology	Key Findings
Chiu, Moorhouse, et al. (2024)	Teacher support and student motivation to learn with AI	Experimental study	Finds that strong teacher support can boost student motivation and self-efficacy when using AI tools.
Kong et al. (2023)	Evaluating an AI literacy program for senior secondary students	Mixed method	Shows that a well-designed program can enhance students' self-efficacy and ethical awareness.
Sanusi, Olaleye, Agbo, and Chiu (2022)	Learners' competencies in AI education	Quantitative Study	Discusses a range of learner competencies, including attitudes and self-efficacy toward AI learning.
Zhao et al. (2022)	AI literacy for primary and middle school teachers in China	Survey study	Examines teachers' AI literacy, with a focus on their self-efficacy and readiness to teach AI.

Table 6
Six research focused on AI ethical awareness, 2021-2025.

Author (Year)	Study Focus	Methodology	Key Findings
Adams et al. (2023)	Ethical principles for AI in K–12 education	Scoping review	Proposes a set of core ethical principles, like fairness and transparency, to guide AI integration in schools.
Ma et al. (2025)	Fostering responsible AI literacy	Systematic review	Synthesizes research on K–12 AI ethics education, highlighting key themes and pedagogical strategies.
Ng et al. (2024)	AI literacy education in secondary schools	Systematic Review	Discusses various aspects of AI literacy, with ethical awareness highlighted as a core component.
Tagare et al. (2025)	K–12 teachers' ethical competencies for AI literacy	Systematic review	Examines the ethical competencies teachers need to develop to effectively teach AI ethics.
Williams et al. (2023)	AI + Ethics curricula for middle school youth	Mixed-methods approach	Provides insights into how ethical curricula can be integrated with technical AI learning.
Zhang et al. (2023)	Integrating ethics with technical learning for middle school students	Exploratory study	Explores a curriculum model that promotes AI literacy by linking technical learning with discussions on ethics and career futures.

(2023), which proposes foundational ethical principles for AI in K–12 education.

Together, these studies highlight that the final component of K–12 AI competency involves an awakening of a student's metacognitive and ethical capacities. It's about building a generation that not only understands and uses AI but also feels empowered and ethically prepared to shape its future. The literature here underscores that true AI literacy extends beyond technical knowledge and practical and cognitive skills, encompassing the critical and humanistic aspects of engaging with AI.

After conducting a systematic review and analysis of 54 research articles, we identified two key gaps in the current literature on AI competency in education. First, although many studies have discussed understanding, using, and developing higher-order thinking with AI, these aspects are often treated as separate entities. There is little discussion of how understanding AI might shape the way people use it, or how using AI could, in turn, help students develop deeper thinking and judgment. Second, when it comes to higher-order competencies, most research focuses on ethical awareness in terms of making the right decisions. However, few studies explore how students build their inner values, self-awareness, and spiritual strength, the deeper foundations that help them stay grounded as human while working with AI.

5. A framework proposed for AI competency in school education

To address the challenges outlined above and ensure the responsible integration of generative AI into school education, an iterative pedagogical approach is essential in K–12 education. Such a framework should empower students to develop self-efficacy in understanding and using AI, enhance their metacognitive skills for learning with AI, and, most importantly, recognize and value their unique human qualities. Rather than relying heavily on AI, students should be guided to master it by cultivating a mindset of pure simplicity, ensuring they remain in control of the tools they use.

As Fig. 2 presents, we propose a framework consisting of three interrelated dimensions: Understanding AI, Using AI, and Unleashing Human Potential in working with AI (1) The Understanding Dimension emphasizes students' conceptual understanding of AI. (2) The Using Dimension focuses on the pedagogical adoption of AI tools to enhance learning. (3) The Unleashing Dimension highlights the development of unique value of human qualities.

This framework conceptualizes Understanding, Using, and Unleashing not as isolated or sequential components, but as an interconnected and cyclical process of learning and growth. A solid understanding of AI—its mechanisms, strengths, and limitations—enables students to use AI tools more strategically, critically, and responsibly. In turn, active and reflective use allows learners to deepen their conceptual understanding, as they begin to see how AI functions, where it succeeds, and where it falls short in real learning contexts. Through this iterative interaction, meaningful use also becomes a pathway to unleashing human potential. As students engage with AI as cognitive partners, they cultivate self-efficacy, metacognitive awareness, and ethical judgment — the higher-order capacities that define human learning. Yet this process does not end there: as these human capacities mature, they in turn reshape how students use AI — transforming their engagement from functional assistance to creative collaboration. In this way, Understanding, Using, and Unleashing form a dynamic loop that continually strengthens both technological literacy and human growth. The goal of AI education, therefore, is not only to teach students how to use AI, but to help them understand, transform, and transcend those interactions in ways that affirm their humanity.

Theoretically, this framework reaffirms the importance of placing human values at the centre of AI education. In existing AI ethics literature, human autonomy is recognized as a key principle—it is human who should decide whether and how to delegate decisions to AI systems in pursuit of human-chosen goals (Floridi & Cowls, 2022; Floridi et al., 2018; Kong & Zhu, 2025). Autonomy can be promoted through transparency, informed consent, and improved AI literacy (Jobin et al., 2019), yet its significance extends beyond regulation. Drawing on psychological perspectives such as self-determination theory, autonomy reflects a person's sense of volition and self-endorsed action (Ryan & Deci, 2020), which is essential for motivation, growth, and well-being (Calvo et al., 2020). From this standpoint, our framework highlights that developing AI competency is not only about ethical or technical operation, but about fostering human autonomy, agency, and inner development—helping students learn to work with AI in ways that affirm their values, critical judgment, and sense of self.

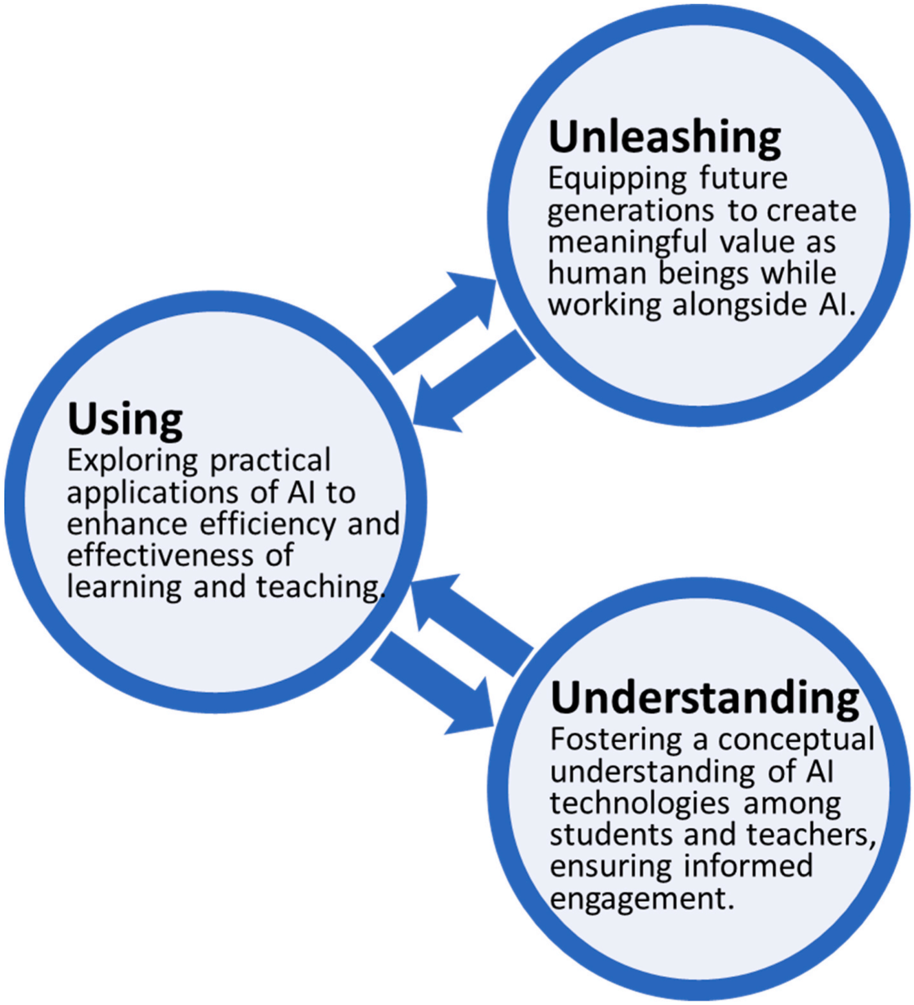

Fig. 2. The framework of understanding AI, using AI and unleashing human potential.

5.1. *Understanding AI: conceptual understanding of AI*

A foundational dimension of AI Competency lies in cultivating conceptual understanding—grasping what AI is, how it works, and what it is not. Two key issues emerge in this area: first, the allure of anthropomorphism, where learners misinterpret AI as sentient or human-like; and second, the need to reclaim conceptual understanding through education by helping students build accurate expectations and learn how to collaborate with AI meaningfully under adult guidance.

5.1.1. *The allure of anthropomorphism*

The story of Hans the Horse—often referred to as “Clever Hans”—offers a revealing analogy for how we sometimes misinterpret artificial intelligence. In the early 20th century, Hans amazed audiences by appearing to perform arithmetic, tap out answers to complex questions, and respond to prompts with remarkable accuracy. Many believed the horse possessed human-like intelligence. However, a scientific investigation by Oskar Pfungst revealed that Hans was not actually solving problems or understanding language. Instead, he was picking up on subtle, unintentional cues from his human trainer—changes in posture, facial tension, or breathing—that signalled when to stop tapping his hoof (Pfungst, 1911). What seemed like understanding was a form of conditioned response to human behaviour. This story parallels our tendency today to over-attribute intelligence or comprehension to AI systems. When ChatGPT composes a poem or when an image generator creates a surreal artwork, we might be tempted to view these outputs as proof of genuine creativity or understanding. But much like Hans, the machine is responding to statistical patterns in data and human inputs—it is not “thinking” in the human sense. A similar cautionary lesson comes from the realm of computer vision, specifically through the phenomenon of adversarial images examples. These are images that have been subtly altered in ways imperceptible to the human eye yet cause AI systems to misclassify them entirely. For instance, a panda image can be modified with barely noticeable noise, leading an AI to confidently misidentify it as a gibbon (Goodfellow et al., 2014). While the image still looks the same to us, the AI's interpretation drastically changes. This again reveals that AI does not “see” the way we do; it processes information through high-dimensional patterns that are alien to human perception.

Both examples—Clever Hans and adversarial images—illuminate a critical truth: machines and human operate on fundamentally different epistemic grounds. While human infer meaning through context, emotion, and embodied experience, machines rely on statistical correlations and computational cues. What may appear as shared understanding is often an illusion, a convergence of behaviour that masks profound internal divergence.

However, anthropomorphic interpretations of AI are now alarmingly widespread just as Table 7 shows. Technology becomes seductive when its affordances align with our emotional vulnerabilities (Turkle, 2011). This tendency is further reinforced by AI systems’ human-like interfaces and natural language outputs, which create an illusion of understanding, empathy, or agency. In such a context, without proper education or critical literacy, the public is easily misled into believing that AI systems possess human-like consciousness or moral intentionality.

Table 7
Examples of anthropomorphic interpretations of AI.

Term	Common Misinterpretation	Actual Meaning in AI Context
Temperature	AI feels "hot" or "cold"; emotional nuance	A parameter that controls randomness in generated responses
Creativity	AI can invent like a human artist or thinker	Recombination of patterns from training data; no intentional creation
Hallucination	AI is "seeing things" like a confused mind	Outputting plausible but factually incorrect or fabricated content

Moreover, the anthropomorphizing of AI often goes hand in hand with the erasure of the vast human labour that underpins its development. AI is neither autonomous nor intelligent in any human sense—it is a profoundly material system, grounded in natural resource extraction, computational infrastructures, invisible labour, and long-standing regimes of classification (Crawford, 2021) Without such intensive human and material support, AI systems would not 'understand' or 'recognize' anything at all.

This widespread misconception—that AI systems possess human-like cognition or moral agency—has significant societal consequences. When AI is seen as an autonomous agent rather than a sociotechnical construct, responsibility becomes dangerously abstracted. Ethical failures or algorithmic harms are more easily attributed to the "black box" of AI rather than to the institutions, developers, and power structures behind it. This obscures accountability, allowing corporations and governments to deflect scrutiny by invoking technological inevitability or neutrality. Without widespread critical AI literacy, the public remains vulnerable not only to manipulation and misinformation, but also to systemic injustices embedded within these technologies. In this sense, the myth of autonomous intelligence is not just misleading—it is politically consequential.

5.1.2. Reclaiming understanding through school education

Students need to grasp the basic concepts and operational principles behind AI, especially the core ideas of machine learning, deep learning, and how algorithms function. AI education should begin with intuitive, age-appropriate explanations of how AI works—for instance, introducing the logic behind supervised learning (such as linear regression), unsupervised learning (like K-means clustering), and neural networks. Students should also understand how these algorithms are trained on large datasets and applied to solve real-world problems, such as image recognition, text generation, or medical diagnostics (Marques et al., 2020).

Importantly, students do not need to master complex mathematical formulas nor become proficient programmers to build AI literacy. However, a basic understanding of key mathematical concepts—such as mean, standard deviation, slope, and Euclidean distance—can help them follow how AI systems process and interpret data (Kong et al., 2023). Teaching AI in this accessible and interdisciplinary way lowers the barrier to entry and makes AI literacy a feasible goal for all learners, not just those in STEM tracks (Kong et al., 2023; Kong & Hu, 2025; Long & Magerko, 2020). It also prepares students—and future citizens—to critically evaluate and meaningfully participate in an AI-saturated world (Kong et al., 2024).

Beyond technical knowledge of how algorithms operate, students should also understand the material and human foundations of AI systems. AI is not an autonomous or immaterial force—it is a sociotechnical system built on the extraction of natural resources, the use of energy-intensive computational infrastructure, and the often-invisible labour of thousands of workers who annotate data, manage servers, and maintain digital platforms.

To achieve this, education systems must provide sustained institutional support. This includes integrating AI literacy into national curricula, designing cross-disciplinary teaching materials, and offering professional development opportunities for teachers. Educators themselves need training to feel confident in teaching both the technical and social dimensions of AI. A meaningful and inclusive AI education policy should treat AI literacy not as a luxury, but as a foundational component of twenty-first-century citizenship.

5.2. Using AI: critical and reflective AI use in learning and teaching

Building on clear understanding of AI's strengths, limitations, and underlying mechanisms, students and teachers need to engage AI in diverse and meaningful ways: through strategic use such as prompt design as cognitive scaffolding; comparative use by interrogating AI outputs against human reasoning; reflective use by learning from AI's failures and misjudgements; and the cultivation of a critical pedagogical culture that supports responsible and scalable integration of AI in education.

5.2.1. Strategic use: prompt design as cognitive scaffolding

One of the most foundational skills in using generative AI is prompt design. For example, vague prompts like "Explain climate change" may yield surface-level summaries, whereas more precise prompts—such as "Compare the causes of climate change in developed and developing countries with examples from 2000 to 2020"—encourage more detailed, contextualized outputs. In this sense, prompt design becomes a form of cognitive scaffolding: students externalize their thought processes through language, and the AI's responses help clarify, expand, or even challenge those thoughts.

For teachers, strategic prompting can support differentiated instruction. They can tailor tasks for different student profiles—asking AI to simplify a complex concept for younger learners, or to simulate a debate between opposing historical figures for more advanced students. Teachers can also use AI to generate alternative examples, analogies, or counterarguments, encouraging students to think beyond binary perspectives. The key is not to let AI dominate the instructional process, but to use it as a flexible pedagogical tool that enhances students' critical engagement with content.

5.2.2. Comparative use: interrogating with the machine

Another crucial layer is comparative AI use. Students should be encouraged to consult multiple platforms—such as ChatGPT, Claude, Gemini, or specialized subject-based AIs—and compare how each system interprets the same question. This practice reveals the underlying design logic, training data, and bias of each tool. For example, a literature student might ask two AI models to analyse a poem and notice that one emphasizes formal structure while the other highlights thematic motifs. These discrepancies provide openings for meta-level discussion: Why do the answers differ? What assumptions or cultural frames might the AI be encoding? Such exercises cultivate interrogative use, where students treat AI not as an oracle, but as a discursive participant—one whose statements are always contingent, partial, and shaped by architecture. By identifying patterns in variation, learners sharpen their capacity for synthesis and judgment.

5.2.3. Reflective use: learning from AI's failures

When students encounter hallucinated citations, factual inaccuracies, culturally biased language, or vague generalizations, they are prompted to question the system's limitations. These "failures" are not just technical bugs—they are pedagogical opportunities. Teachers can ask: Why did the AI produce this answer? What assumptions did it make? What data might it be missing? Such reflection nurtures epistemic humility and highlights the statistical, rather than sentient, nature of generative models. Moreover, students learn to reframe AI responses as provisional knowledge—a starting point for further research, not a final answer. This encourages iterative thinking. A weak response leads to prompt revision; an incomplete answer sparks supplementary questions; a biased claim invites counter examples. Over time, students internalize a habit of recursive inquiry, in which the AI is neither master nor mirror, but a responsive environment for exploration.

5.2.4. Pedagogical culture: supporting critical use at scale

For this vision to be realized, schools must foster a pedagogical culture that values exploration over efficiency, and critique over convenience. Teachers need institutional support, professional development, and time to experiment with AI in low-stakes settings. (Kong, 2025; Kong & Yang, 2024, 2025). Assessment designs can reward students not just for answers, but for showing how they iterated through prompts, revised based on output, or reflected on bias. A truly "AI-ready" classroom is not one that uses AI frequently, but one where students and teachers alike are empowered to ask: Why this tool, in this moment, for this purpose?

5.3. Unleashing potential of human: creating meaningful value as human beings while working alongside with AI

Beyond using AI tools effectively, education must aim to unleash students' deeper potential by encouraging them to see AI not as a replacement, but as a partner in intellectual growth. This requires recognizing and cultivating uniquely human capacities that cannot be outsourced to machines—self-efficacy for learning, metacognitive awareness, and a spiritual self-capable of remaining grounded in values amidst technological advancement.

The evolution of human potential follows a remarkable trajectory. According to Harari (2024) in *Nexus*, the dominance of *Homo sapiens* stems not from exceptional individual intelligence, but from the unique ability to create and sustain intersubjective fictions—shared beliefs or narratives, such as religions, ideologies, and cultural myths—that enable strangers to cooperate on unprecedented scales. These constructs transcend immediate physical needs, fostering collective purpose, cultural complexity, and introspective pursuits such as ethical inquiry and spiritual reflection. Ultimately, it is humanity's distinctive capacity to harness information not only for survival, but also for intellectual and spiritual enlightenment that defines its progress. Today, the rise of artificial intelligence (AI) marks yet another turning point in this trajectory. In many ways, AI could be seen as the next "shared construct," reshaping human cooperation and redefining societal purpose. This moment calls for a deeper consideration of how human values and achievements can continue to shape the future in the face of transformative technological change.

5.3.1. Cultivating the spiritual self in an AI era

To truly unleash our potential, Fig. 3 shows a threefold pathway to achieve the learning goal in K–12: (1) Informed passion: Self-efficacy for learning; (2) Reflection: Metacognitive capacities; (3) Spiritual Self: Pure simplicity.

5.3.2. Informed passion: self efficacy for learning

In the age of generative AI and beyond, the ability to think—not just to retrieve—becomes a foundational human strength. This is closely tied to a learner's self-efficacy, which is their belief in their capability to organise and execute a course of action to manage a given situation (Bandura, 1997). Informed passion refers to a deep, purposeful enthusiasm that is guided by knowledge and critical thinking. It is not just raw interest or excitement, but a deliberate and informed drive to engage, explore, and create meaning in a focused area.

The presence of AI in the classroom introduces new sources for building self-efficacy. Consider a student who is struggling to structure an argument for a history paper. They have the facts but lack the confidence to craft a coherent thesis and supporting points. They decide to use a generative AI tool as a thinking partner. The student prompts the AI with their raw ideas, and the AI provides several structured outlines. The student doesn't copy an outline but instead selects the one that best fits their own perspective and then refines it, adding their unique voice and evidence. When the final paper receives positive feedback, the student's success is directly attributable to their own critical judgement and curation—not the AI's generation. Their self-efficacy grows, fostering the confidence that they can tackle future, more challenging writing tasks not by relying on AI, but by leveraging it as a tool to amplify their own intellectual capabilities. This process exemplifies informed passion—a purposeful and knowledge-driven engagement that empowers learners to think critically, act confidently, and grow stronger in the face of new challenges.

5.3.3. Reflection: metacognitive capacities

Self-regulated learning (SRL) is the process that allows learners to succeed and cultivate intellectual independence. SRL theory, particularly as defined by Zimmerman (2000), outlines a cyclical process that is highly applicable to learning with AI. As a cornerstone of SRL, metacognition—or 'thinking about one's thinking'—is the ability to plan, monitor, and assess one's own learning. This capacity is essential for preventing over-reliance on AI. For example, consider a student using an AI-powered maths tutor. A student lacking self-regulatory skills might simply accept the AI's explanation and move on without truly understanding the underlying concept. A student who is actively regulating their learning, however, would engage in a reflective process. In the forethought phase, they might plan how to approach a challenging problem using the AI. During the performance phase, they would actively monitor their own comprehension, pausing to ask themselves, 'Does this explanation truly make sense to me?' and 'Am I able to replicate this process on my own?' Finally, in the self-reflection phase, they would assess their learning, perhaps by attempting a similar problem without AI assistance to test their understanding. This reflective process allows the student to verify actively the AI's logic, identify gaps in their knowledge, and correct misunderstandings, ultimately reinforcing their learning and building a deeper understanding of the subject matter.

5.3.4. Spiritual self: A state of pure simplicity

The final component of human growth in an AI-saturated age is not primarily technical but existential: the cultivation of the spiritual self. Spiritual self-cultivation involves maintaining a clear understanding of the intrinsic values that guide one's actions when working with AI and doing good. It emphasizes ensuring that these value-based decisions remain inherently human, rather than delegating them to AI systems,

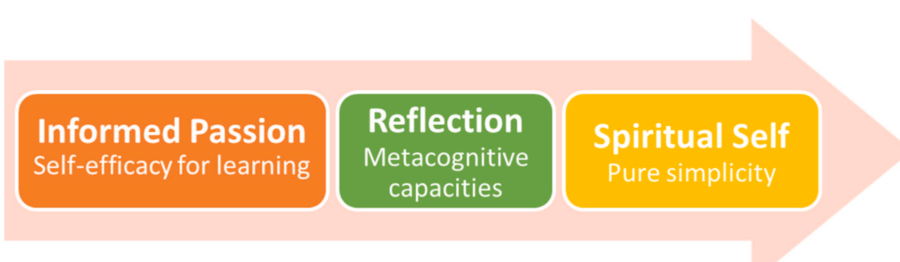

Fig. 3. Unleashing the potential of human.

which lack the capacity to make such judgments.

This turn is not without precedent. Educational thinkers from Kant (1982), Newman (1852), and Dewey (1916) to the Confucian tradition of self-cultivation (Li, 2012) have long emphasized that education ultimately concerns the formation of the self rather than the mere transmission of skills. Contemporary philosophers extend this lineage by insisting that education's intrinsic task is subjectivation (Biesta, 2009) and that valuation and student self-formation remain central to educational practice (Marginson, 2024). In the context of an AI-saturated age—where technical competence is increasingly automated—these perspectives signal an urgent need to reorient education toward existential growth, underscoring that the cultivation of the spiritual self is not optional but essential.

We argue that a self anchored in pure simplicity (Legge, 1891). This state of pure simplicity is endorsed in the philosophical wisdom found in classical traditions. In Zhuangzi (《莊子》), it is written: 「有機械者必有機事,有機事者必有機心。機心存於胸中,則純白不備; 純白不備,則神生不定,神生不定者,道之所不載也。」 "Where you have machines, then you get certain kinds of problems; where you get certain kinds of problems, then you find a heart warped by these problems. Where you get a heart warped, its purity and simplicity are disturbed. When purity and simplicity are disturbed, then the spirit is alarmed, and an alarmed spirit is no place for the Tao to dwell." (Zhuangzi, ca. 4th century BCE/Palmer et al., Trans., 2006). Scheming mind is a mindset that takes advantage of environment for self-interest. Pure simplicity is not taking advantage of environment for self-interest. In an age of endless prompts and limitless AI outputs, human is struggling with the scheming mind and pure simplicity. The spiritual self is the one who knows when to pause, when to disengage, and when to be still and does good. It resists the automation of attention. Fu (2023) quotes Zhuangzi's warning that "Where you have machines, then you get certain kinds of problems; where you get certain kinds of problems, then you find a heart warped by these problems." He argues that "this perspective urges us to rethink our relationship with technology in the contemporary world. While we cannot turn back the clock, we can strive to minimize the influence of the machines, ensuring it does not permeate every aspect of our lives" (pp. 174-175).

While Fu's argument highlights the potential dangers of over-reliance on technology and its capacity to foster manipulation, he does not provide specific guidance on how individuals can collaborate with technology in a way that prevents the influence of a scheming mind on the cultivation of the spiritual self. We argue that coexisting with technology does not necessarily entail the inevitable presence of a scheming or self-serving mindset. By intentionally cultivating a settled spirit in our actions and decisions, individuals can resist the influence of manipulative tendencies and engage with technology in a manner that aligns with their core values and do good, fostering pure simplicity in both thought and action.

For instance, consider a secondary student who, after acquiring knowledge about AI, decides to teach these skills to middle school students or elderly individuals who need them. Rather than using AI solely to complete tasks more easily for personal gain, this student focuses on helping others bridge the technology gap, empowering them to navigate an increasingly digital world. This act of sharing knowledge not only reduces inequities in access to technology but also helps the student develop a stronger value system. By reflecting on the societal role of technology and their responsibility to use it ethically, the student begins to reimagine how technology should exist within society—not as a tool of convenience or exploitation, but as a means of fostering connection, understanding, and equity. To cultivate the spiritual self is not to reject technology but to transcend its control—to engage with it thoughtfully and purposefully, ensuring that its use reflects and reinforces one's core values.

6. Conclusion, implication and limitation

This study conducted a scoping review of 54 peer-reviewed publications to map how existing research has conceptualised students' AI competencies across three key dimensions: Understanding AI, Using AI, and Unleashing human potential in working with AI. Although current studies discuss these dimensions extensively, they do not put them together for the goal of developing human values in deploying the powerful technology of AI. Drawing from this analysis, we propose a conceptual framework for developing AI competency that highlights human values. The proposed framework consists of three interrelated dimensions: understanding AI for informed engagement, reflective use of AI for metacognitive development, and growth of spiritual self through vitalising a pure simplicity mindset for unleashing students' potential in working with AI. There are three implications for implementing this framework in K–12 education.

Firstly, this framework underscores the importance of providing opportunities for students to understand how AI works and, without anthropomorphizing it, enabling them to approach AI tools as resources for enhancing their informed passion for self-efficacy development. Building on Bandura's (1997) theory, it highlights the importance of fostering students' confidence in their ability to tackle increasingly complex learning tasks. This confidence is not rooted in dependence on AI but in leveraging AI as a tool to amplify their intellectual capabilities. By nurturing self-efficacy, students are empowered to view AI not as a crutch but as an enabler of their own growth and potential. This could guide K–12 educators in designing learning experiences that build students' confidence in using AI effectively and wisely, derived from an understanding of how AI works, rather than passively relying on these tools.

Secondly, this framework provides guidance for teachers to support students in developing metacognitive skills through reflective use of AI tools. The framework underscores the pivotal role of self-regulated learning (Zimmerman, 2000), which empowers students to "learn how to learn" in the context of AI integration. Drawing on Flavell's (1979) seminal work, metacognition refers to both awareness of one's cognitive processes and the ability to regulate them, including planning, monitoring, evaluation, and strategic adjustment. In human-AI interactions, students are required to engage in deliberate metacognitive practices. For example, they must critically assess the reliability and accuracy of AI-generated outputs, reflect on the effectiveness of their prompt engineering, identify potential biases in AI outputs, and iteratively refine their strategies to achieve optimal learning outcomes. For K–12 education, this means equipping students with the ability to plan, monitor, and reflect on their learning processes in their interactions with AI tools, to ensure they maintain control and agency over their learning experiences.

Thirdly, this framework highlights the need to cultivate students with a pure simplicity mindset. In other words, students should not always take advantage of their environment for self-interest without their spiritual self knowing when to pause, when to disengage, and when to be still in working with AI tools. This perspective aligns with contemporary educational priorities (Miao & Holmes, 2023; UNESCO, 2023) and recognises the risks of over-reliance on AI, such as the erosion of ethical responsibility and critical autonomy (Miao & Holmes, 2023; UNESCO, 2023; Vartiainen et al., 2024). To address these concerns, students must cultivate not only technical proficiency but also critical reflection and value-oriented thinking rooted in humanistic traditions. By doing so, they can navigate the challenges when using AI in learning whilst preserving their autonomy, ultimately ensuring that AI serves as a tool to enhance—not diminish—human potential. We need to cultivate students to value themselves as humans and not as slaves of AI.

These implications offer guidance for K–12 educators to consider both the ethical use of AI as well as fully recognizing the need to develop the spiritual self of students in integrating AI for pedagogical use. Teachers should foster students' ability to make ethical and value-driven decisions when engaging with AI tools. This framework offers actionable guidance for curriculum design and teacher development in AI in K–12 education.

Despite these contributions, the study has two limitations. First, as is inherent to the scoping review, the analysis of this study is constrained by the 54 peer-reviewed publications. Second, whilst the proposed framework offers a theoretical model, it has not yet undergone empirical validation. Future research should therefore be conducted to test the practical feasibility and effectiveness of this AI competency framework in real classroom settings. Additionally, given the framework's emphasis on unleashing human potential, future work should explore how to measure and assess students in the context of using AI in education.

CRediT authorship contribution statement

Siu Cheung Kong: Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Funding acquisition, Conceptualization. **Wenxi Hu:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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